Handwritten digits classification using neural network

In this notebook we will classify handwritten digits using a simple neural network which has only input and output layers. We will than add a hidden layer and see how the performance of the model improves

In [1]:
```
import tensorflow as tf
from tensorflow import keras
import matplotlib.pyplot as plt
%matplotlib inline
import numpy as np
```

```
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages
py\_distributor_init.py:30: UserWarning: loaded more than 1 DLL from .libs:
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages\num
py\.libs\libopenblas.EL2C6PLE4ZYW3ECEVIV3OXXGRN2NRFM2.gfortran-win_amd64.dll
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages\num
py\.libs\libopenblas64__v0.3.21-gcc_10_3_0.dll
  warnings.warn("loaded more than 1 DLL from .libs:"
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages\sci
py\__init__.py:146: UserWarning: A NumPy version >=1.16.5 and <1.23.0 is requir
ed for this version of SciPy (detected version 1.24.3
  warnings.warn(f"A NumPy version >={np_minversion} and <{np_maxversion}"
```

In [3]:
```
(X_train, y_train) , (X_test, y_test) = keras.datasets.mnist.load_data()
```

In [4]:
```
len(X_train)
```

Out[4]:
```
60000
```

In [5]:
```
len(X_test)
```

Out[5]:
```
10000
```

In [6]:
```
X_train[0].shape
```

Out[6]:
```
(28, 28)
```

In [7]:
```
X_train[0]
```

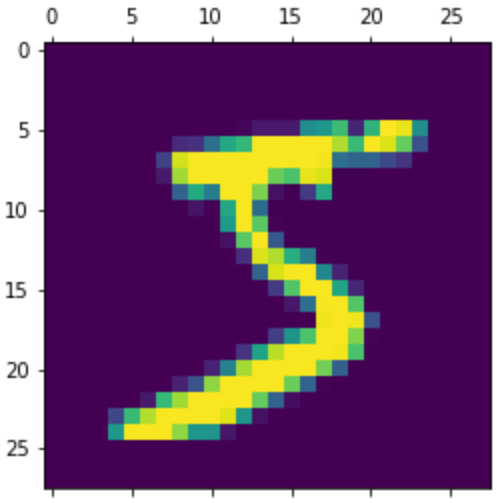
Out[7]:
```
array([[  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   3,
         18,  18,  18, 126, 136, 175,  26, 166, 255, 247, 127,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,  30,  36,  94, 154, 170,
        253, 253, 253, 253, 253, 225, 172, 253, 242, 195,  64,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,  49, 238, 253, 253, 253, 253,
        253, 253, 253, 253, 251,  93,  82,  82,  56,  39,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,  18, 219, 253, 253, 253, 253,
        253, 198, 182, 247, 241,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,  80, 156, 107, 253, 253,
        205,  11,   0,  43, 154,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,  14,   1, 154, 253,
         90,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0, 139, 253,
        190,   2,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,  11, 190,
        253,  70,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,  35,
        241, 225, 160, 108,   1,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
         81, 240, 253, 253, 119,  25,   0,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,  45, 186, 253, 253, 150,  27,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,  16,  93, 252, 253, 187,   0,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0, 249, 253, 249,  64,   0,   0,   0,   0,   0,
          0,   0],
       [  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,  46, 130, 183, 253, 253, 207,   2,   0,   0,   0,   0,   0,
          0,   0],
```

```
[  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,  39,
 148, 229, 253, 253, 253, 250, 182,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,  24, 114, 221,
 253, 253, 253, 253, 201,  78,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,   0,   0,   0,   0,  23,  66, 213, 253, 253,
 253, 253, 198,  81,   2,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,   0,   0,  18, 171, 219, 253, 253, 253, 253,
 195,  80,   9,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,  55, 172, 226, 253, 253, 253, 253, 244, 133,
  11,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0, 136, 253, 253, 253, 212, 135, 132,  16,   0,
   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0],
[  0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
   0,   0]], dtype=uint8)
```

In [9]:

```
plt.matshow(X_train[0])
```

Out[9]: <matplotlib.image.AxesImage at 0x1fe79cb99e8>



In [10]:

```
y_train[0]
```

Out[10]: 5

In [11]:
```
X_train = X_train / 255
X_test = X_test / 255
```

In [12]:
```
X_train[0]
```

```
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.97647059, 0.99215686, 0.97647059,
       0.25098039, 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        ],
      [0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.18039216,
       0.50980392, 0.71764706, 0.99215686, 0.99215686, 0.81176471,
       0.00784314, 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        ],
      [0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.15294118, 0.58039216, 0.89803922,
       0.99215686, 0.99215686, 0.99215686, 0.98039216, 0.71372549,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        ],
      [0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.09411765, 0.44705882, 0.86666667, 0.99215686, 0.99215686,
       0.99215686, 0.99215686, 0.78823529, 0.30588235, 0.        ,
```

In [13]:
```
X_train_flattened = X_train.reshape(len(X_train), 28*28)
X_test_flattened = X_test.reshape(len(X_test), 28*28)
```

In [14]:
```
X_train_flattened.shape
```

Out[14]:
```
(60000, 784)
```

In [15]:
```
X_train_flattened[0].        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.
```

Out[15]:
```
array([0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
       0.        , 0.        , 0.        , 0.        , 0.        ,
```

Very simple neural network with no hidden layers

In [45]:
```
model = keras.Sequential([
    keras.layers.Dense(10, input_shape=(784,), activation='sigmoid')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(X_train_flattened, y_train, epochs=5)
```

```
Epoch 1/5
1875/1875 [==============================] - 3s 1ms/step - loss: 0.4886 - accur
acy: 0.8775
Epoch 2/5
1875/1875 [==============================] - 3s 1ms/step - loss: 0.3060 - accur
acy: 0.9156
Epoch 3/5
1875/1875 [==============================] - 2s 1ms/step - loss: 0.2848 - accur
acy: 0.9214
Epoch 4/5
1875/1875 [==============================] - 2s 1ms/step - loss: 0.2747 - accur
acy: 0.9243
Epoch 5/5
1875/1875 [==============================] - 2s 1ms/step - loss: 0.2677 - accur
acy: 0.9262
```

Out[45]:
```
<tensorflow.python.keras.callbacks.History at 0x1fe24f47a90>
```

In [46]:
```
model.evaluate(X_test_flattened, y_test)
```

```
313/313 [==============================] - 0s 985us/step - loss: 0.2670 - accur
acy: 0.9257
```

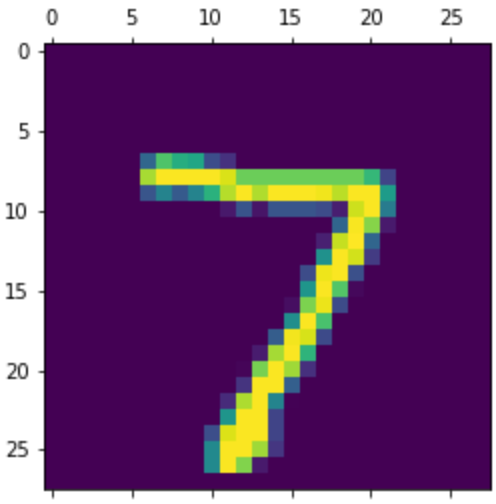
Out[46]:
```
[0.26697656512260437, 0.9257000088691711]
```

In [47]:
```
y_predicted = model.predict(X_test_flattened)
y_predicted[0]
```

Out[47]:
```
array([1.7270680e-05, 1.3593615e-10, 4.5622761e-05, 7.5602829e-03,
       1.3076769e-06, 7.5061922e-05, 1.7646971e-09, 6.9968843e-01,
       7.8440302e-05, 8.1232190e-04], dtype=float32)
```

In [48]:
```
plt.matshow(X_test[0])
```

Out[48]:
```
<matplotlib.image.AxesImage at 0x1fe2322e3c8>
```



np.argmax finds a maximum element from an array and returns the index of it

In [49]:
```
np.argmax(y_predicted[0])
```

Out[49]:
```
7
```

In [50]:
```
y_predicted_labels = [np.argmax(i) for i in y_predicted]
```

In [51]:
```
y_predicted_labels[:5]
```

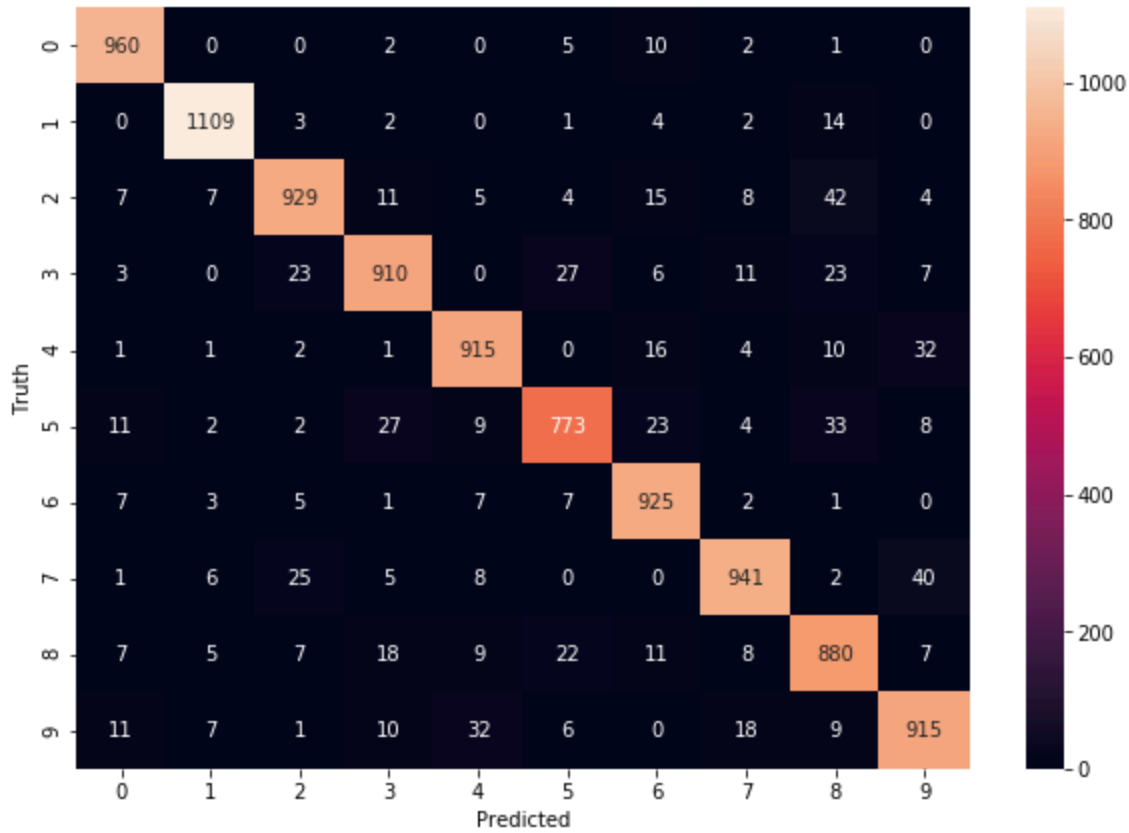
Out[51]:
```
[7, 2, 1, 0, 4]
```

In [52]:
```
cm = tf.math.confusion_matrix(labels=y_test,predictions=y_predicted_labels)
cm
```

Out[52]:
```
<tf.Tensor: shape=(10, 10), dtype=int32, numpy=
array([[ 960,    0,    0,    2,    0,    5,   10,    2,    1,    0],
       [   0, 1109,    3,    2,    0,    1,    4,    2,   14,    0],
       [   7,    7,  929,   11,    5,    4,   15,    8,   42,    4],
       [   3,    0,   23,  910,    0,   27,    6,   11,   23,    7],
       [   1,    1,    2,    1,  915,    0,   16,    4,   10,   32],
       [  11,    2,    2,   27,    9,  773,   23,    4,   33,    8],
       [   7,    3,    5,    1,    7,    7,  925,    2,    1,    0],
       [   1,    6,   25,    5,    8,    0,    0,  941,    2,   40],
       [   7,    5,    7,   18,    9,   22,   11,    8,  880,    7],
       [  11,    7,    1,   10,   32,    6,    0,   18,    9,  915]])>
```

In [53]:
```
import seaborn as sn
plt.figure(figsize = (10,7))
sn.heatmap(cm, annot=True, fmt='d')
plt.xlabel('Predicted')
plt.ylabel('Truth')
```

Out[53]: Text(69.0, 0.5, 'Truth')



Using hidden layer

In [54]:
```
model = keras.Sequential([
    keras.layers.Dense(100, input_shape=(784,), activation='relu'),
    keras.layers.Dense(10, activation='sigmoid')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(X_train_flattened, y_train, epochs=5)
```

```
Epoch 1/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.2925 - accur
acy: 0.9191
Epoch 2/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.1366 - accur
acy: 0.9602
Epoch 3/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0981 - accur
acy: 0.9703
Epoch 4/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0764 - accur
acy: 0.9768
Epoch 5/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0618 - accur
acy: 0.9812
```

Out[54]:
```
<tensorflow.python.keras.callbacks.History at 0x1fe230e7128>
```

In [55]:
```
model.evaluate(X_test_flattened,y_test)
```

```
313/313 [==============================] - 0s 1ms/step - loss: 0.0966 - accurac
y: 0.9716
```

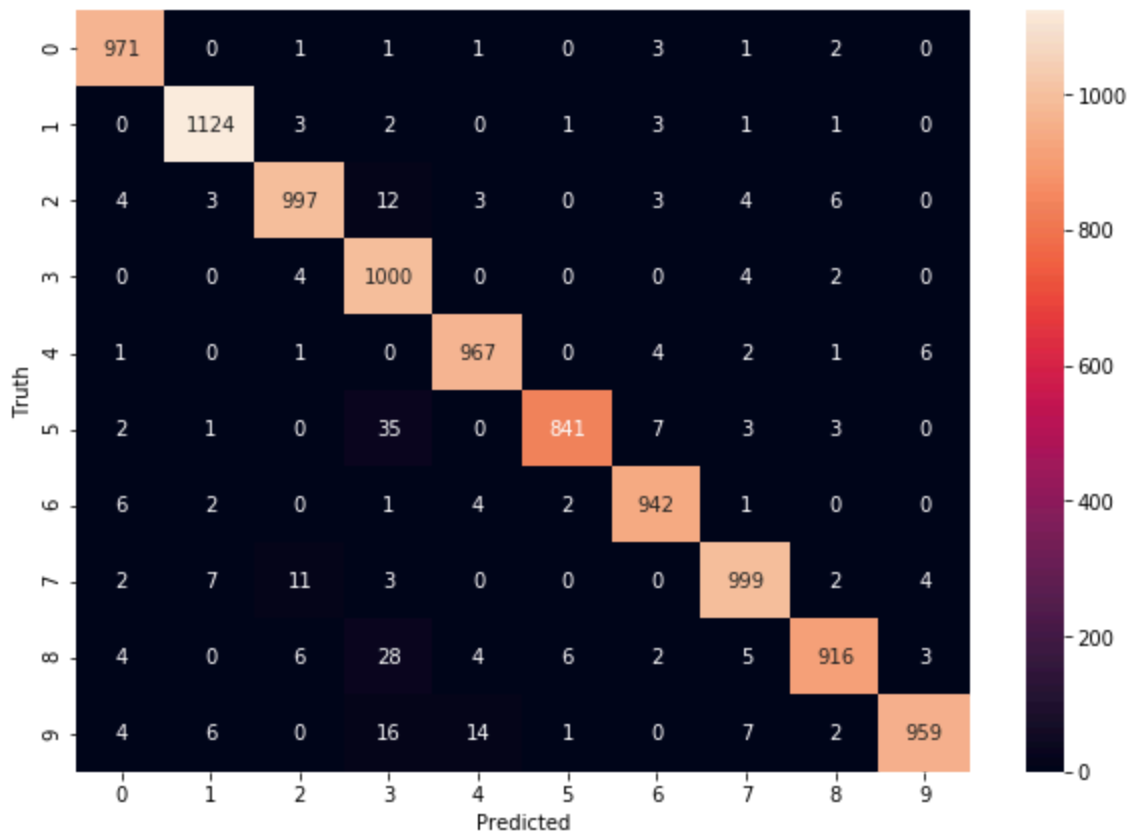
Out[55]:
```
[0.09658893942832947, 0.9715999960899353]
```

In [56]:

```
y_predicted = model.predict(X_test_flattened)
y_predicted_labels = [np.argmax(i) for i in y_predicted]
cm = tf.math.confusion_matrix(labels=y_test,predictions=y_predicted_labels)

plt.figure(figsize = (10,7))
sn.heatmap(cm, annot=True, fmt='d')
plt.xlabel('Predicted')
plt.ylabel('Truth')
```

Out[56]: Text(69.0, 0.5, 'Truth')



Using Flatten layer so that we don't have to call .reshape on input dataset

In [59]:
```
model = keras.Sequential([
    keras.layers.Flatten(input_shape=(28, 28)),
    keras.layers.Dense(100, activation='relu'),
    keras.layers.Dense(10, activation='sigmoid')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(X_train, y_train, epochs=10)
```

```
uracy: 0.9185
Epoch 2/10
1875/1875 [==============================] - 3s 2ms/step - loss: 0.1368 - acc
uracy: 0.9603
Epoch 3/10
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0995 - acc
uracy: 0.9703
Epoch 4/10
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0771 - acc
uracy: 0.9772
Epoch 5/10
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0628 - acc
uracy: 0.9806
Epoch 6/10
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0519 - acc
uracy: 0.9841
Epoch 7/10
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0442 - acc
uracy: 0.9865
Epoch 8/10
```

In [60]:
```
model.evaluate(X_test,y_test)
```

```
313/313 [==============================] - 0s 1ms/step - loss: 0.0813 - accurac
y: 0.9779
```

Out[60]:
```
[0.08133944123983383, 0.9779000282287598]
```